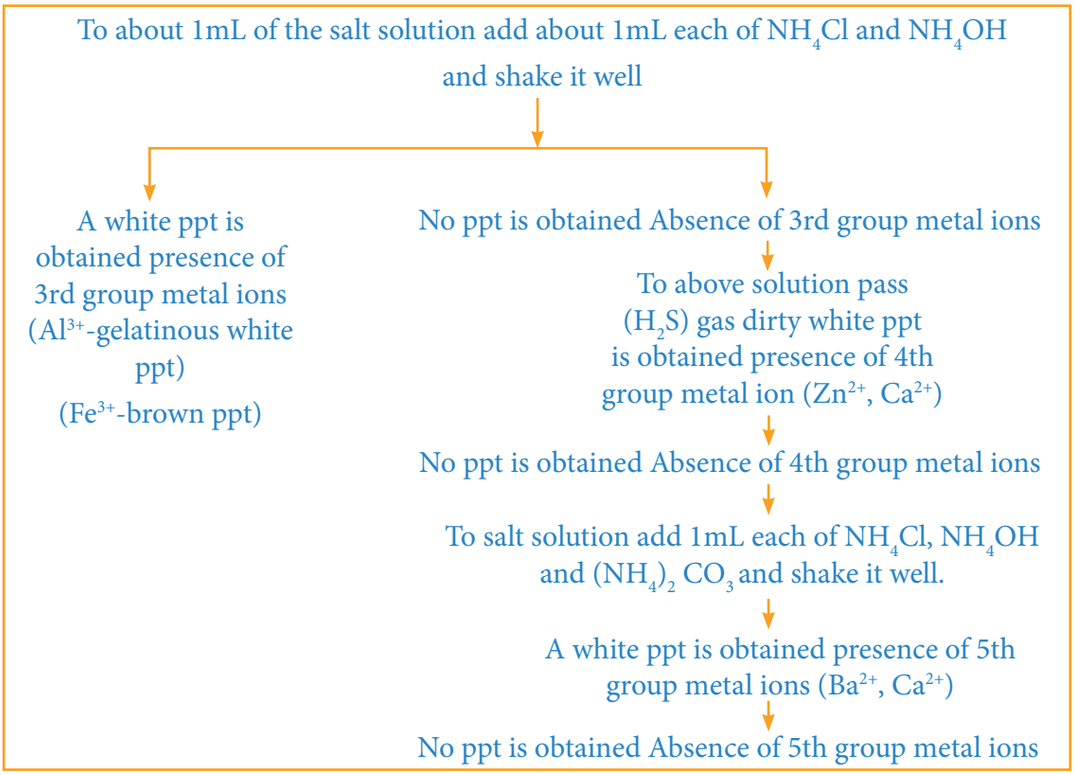
The 3rd group metal ions form metal hydroxide ppt. The 4th group metal ions form metal sulphide ppt.



Analysis of 6 group metal ions
To about 1mL of the original salt solution add about 1mL each of NH_4Cl, NH_4OH and $NH_4H_2PO_4$, and scratch the sides of the test tube. A white ppt is obtained. Presence of magnesium
To about 1mL of the original salt solution add dil. NaOH in drops with shaking. A white ppt insoluble in excess of NaOH is formed. Presence of Magnesium
To about 1mL of the original salt solution add about 1mL of Magneson reagent. A blue ppt is formed. Presence of magnesium.
To about 1mL of the original salt solution add about 1mL each of Nessler's reagent and NaOH. A chocolate brown ppt is obtained. Presence of ammonium

Reasoning:

Magnesium forms $MgNH_4PO_4$ ppt.

Maneson reagent is p-nitro azobenzene resorcinol. The blue ppt is due to precipitation of magneson by $Mg(OH)_2$

Nessler's reagent is prepared by slowly adding potassium iodide to mercury chloride. Initially a white ppt of HgI_2 is obtained. The ppt dissolves in excess forming a clear solution. This clear solution is called the "Nessler's reagent". It is $K_2[HgI_4]$

The bown ppt is due to $HgO.Hg(NH_2)I$. It is a basic mergury (ii) amido amine.

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